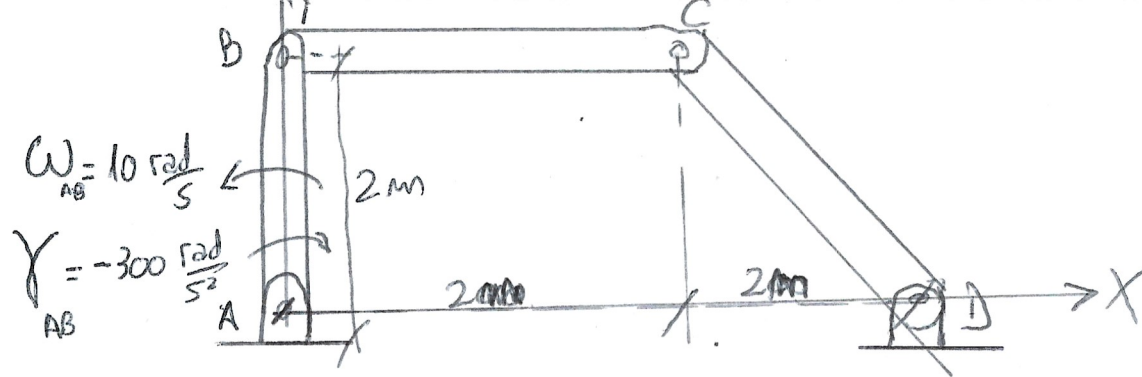




$$A = (0; 0)$$

$$B = (0; 2)$$

$$C = (2; 2)$$

$$D = (4; 0)$$

Empezamos planteando velocidad tangencial en A:

$$\vec{v}_A = \vec{v}_B + \omega_{AB} \times (\vec{A} - \vec{B})$$

$$0 = \vec{v}_B + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 10 \\ 0 & -2 & 0 \end{vmatrix}$$

$$\boxed{\vec{v}_A = 0} \text{ porque está fijo el punto A}$$

$$0 = \vec{v}_B + 20 \hat{j} \Rightarrow \boxed{\vec{v}_B = 20 \hat{j} \frac{m}{s}}$$

$$\vec{v}_B = \vec{v}_C + \omega_{BC} \times (\vec{B} - \vec{C})$$

$$20 \hat{i} = \vec{v}_C + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & \omega_{BC} \\ -2 & 0 & 0 \end{vmatrix}$$

$$20 \hat{i} = \vec{v}_C - 2 \hat{j} \omega_{BC}$$

$$\Rightarrow \boxed{\vec{v}_C = 20 \hat{i} + 2 \hat{j} \omega_{BC}} \quad (1)$$

$$\vec{v}_C = \vec{v}_D + \omega_{CD} \times (\vec{C} - \vec{D})$$

$$\vec{v}_D = 0$$

$$20\vec{i} + 2\vec{j} \omega_{BC} = 0 + \begin{vmatrix} 0 & 0 & \omega_{CD} \\ -2 & 2 & 0 \end{vmatrix}$$

$$20\vec{i} + 2\vec{j} \omega_{BC} = -2\vec{i} \omega_{CD} - 2\vec{j} \omega_{CD}$$

Analizamos en (i) y en (j)

$$(i) \quad 20\vec{i} = -2\vec{i} \omega_{CD} \Rightarrow \boxed{\omega_{CD} = -10 \frac{\text{rad}}{\text{s}}}$$

$$(j) \quad 2\vec{j} \omega_{BC} = -2\vec{j} \omega_{CD} \Rightarrow 2\vec{j} \omega_{BC} = -2\vec{j}(-10)$$

$$\Rightarrow \boxed{\omega_{BC} = 10 \frac{\text{rad}}{\text{s}}} \Rightarrow \text{DE (1)} \quad \boxed{\vec{v}_C = 20\vec{i} + 20\vec{j}}$$

Ahora vamos con las aceleraciones:

$$\boxed{a_A = 0}$$

$$\vec{a}_A = \vec{a}_B + \gamma_{AB} \times (\vec{A} - \vec{B}) + \omega_{AB} \times (\vec{v}_A - \vec{v}_B)$$

$$0 = \vec{a}_B + \begin{vmatrix} 0 & 0 & -300 \\ 0 & -2 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 10 \\ -20 & 0 & 0 \end{vmatrix}$$

$$0 = \vec{a}_B - 600\vec{i} \Rightarrow \boxed{\vec{a}_B = 600\vec{i} \frac{\text{m}}{\text{s}^2}}$$

$$a_B = a_c + \gamma_{bc} X(B-c) + W_{bc} X(N_B - N_c)$$

$$600i = a_c + \begin{vmatrix} i & -j & k \\ 0 & 0 & \gamma_{bc} \\ -2 & 0 & 0 \end{vmatrix} + \begin{vmatrix} i & -j & k \\ 0 & 0 & 10 \\ 0 & -20 & 0 \end{vmatrix}$$

$$600i = a_c - 2j\gamma_{bc} + 200i \Rightarrow a_c = 600i - 200i + 2j\gamma_{bc}$$

$$a_c = 400i + 2j\gamma_{bc}$$

$$a_c = \cancel{a_D} + \gamma_{cd} X(c-d) + W_{cd} X(N_c - N_d)$$

$$400i + 2j\gamma_{bc} = \begin{vmatrix} i & -j & k \\ 0 & 0 & \gamma_{cd} \\ -2 & 2 & 0 \end{vmatrix} + \begin{vmatrix} i & -j & k \\ 0 & 0 & -10 \\ 20 & 20 & 0 \end{vmatrix}$$

$$400i + 2j\gamma_{bc} = -2i\gamma_{cd} - 2j\gamma_{cd} + 200i - 200j$$

Analizamos en (i) y en (j):

$$(i) \quad 400i = -2i\gamma_{cd} + 200i \Rightarrow \gamma_{cd} = -100 \frac{\text{rad}}{\text{s}^2}$$

$$(j) \quad 2j\gamma_{bc} = -2j\gamma_{cd} - 200j \Rightarrow \gamma_{bc} = 0$$